GRADIENT ATOMS: UNSUPERVISED DISCOVERY, ATTRIBUTION AND STEERING OF MODEL BEHAVIORS VIA SPARSE DECOMPOSITION OF TRAINING GRADIENTS

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ABSTRACT

Training data attribution (TDA) methods ask which training documents are responsible for a model behavior. We argue that this per-document framing is fundamentally mismatched to how fine-tuning actually works: models often learn broad concepts shared across many examples. Existing TDA methods are supervised—they require a query behavior, then score every training document against it—making them both expensive and unable to surface behaviors the user did not think to ask about. We present **Gradient Atoms**, an unsupervised method that decomposes per-document training gradients into sparse components ("atoms") via dictionary learning in a preconditioned eigenspace. Among the 500 discovered atoms, the highest-coherence ones recover interpretable task-type behaviors—refusal, arithmetic, yes/no classification, trivia QA—without any behavioral labels. These atoms double as effective steering vectors: applying them as weight-space perturbations produces large, controllable shifts in model behavior (e.g., bulleted-list generation 33% $\rightarrow$ 94%; systematic refusal 50% $\rightarrow$ 0%). The method requires no query–document scoring stage, and scales independently of the number of query behaviors of interest. Code is here: https://github.com/jrosseruk/gradient_atoms

1 INTRODUCTION

When we fine-tune a language model, we want to understand what it has learned. Training data attribution (TDA) is the dominant framework for answering this question: given a model behavior of interest, TDA scores individual training documents by how much each contributed to that behavior (Koh & Liang, 2017; Grosse et al., 2023; Bae et al., 2024).

We argue that this framing has a fundamental problem: **the unit of analysis is wrong**. Fine-tuning does not learn from individual documents in isolation. When a model learns to do arithmetic, that is not because of any single arithmetic example—it is because hundreds of arithmetic examples all push the model's weights in roughly the same direction. The "thing learned" is a shared update direction, not a property of any individual document. Attributing a behavior to document x_{42} is like attributing a river's course to a single raindrop.

This mismatch creates practical problems. Standard TDA is **supervised**: the user must specify a query behavior, then score every training document against it. This requires (1) knowing what behaviors to look for in advance, and (2) an expensive $O(N)$ scoring pass per query. For influence-function methods like EKFAC (Grosse et al., 2023; Bae et al., 2024), attributing Q behaviors costs $O(Q \times N)$ query–document comparisons. If the goal is to understand everything a model learned, this is intractable.

We propose a different question. Instead of "which document caused this behavior?", we ask: **what are the common update directions that clusters of documents jointly induce?** Documents that teach the model the same thing push its weights in the same direction. If we decompose the space of training gradients into sparse components, high-quality components should each isolate a recognisable behavior.

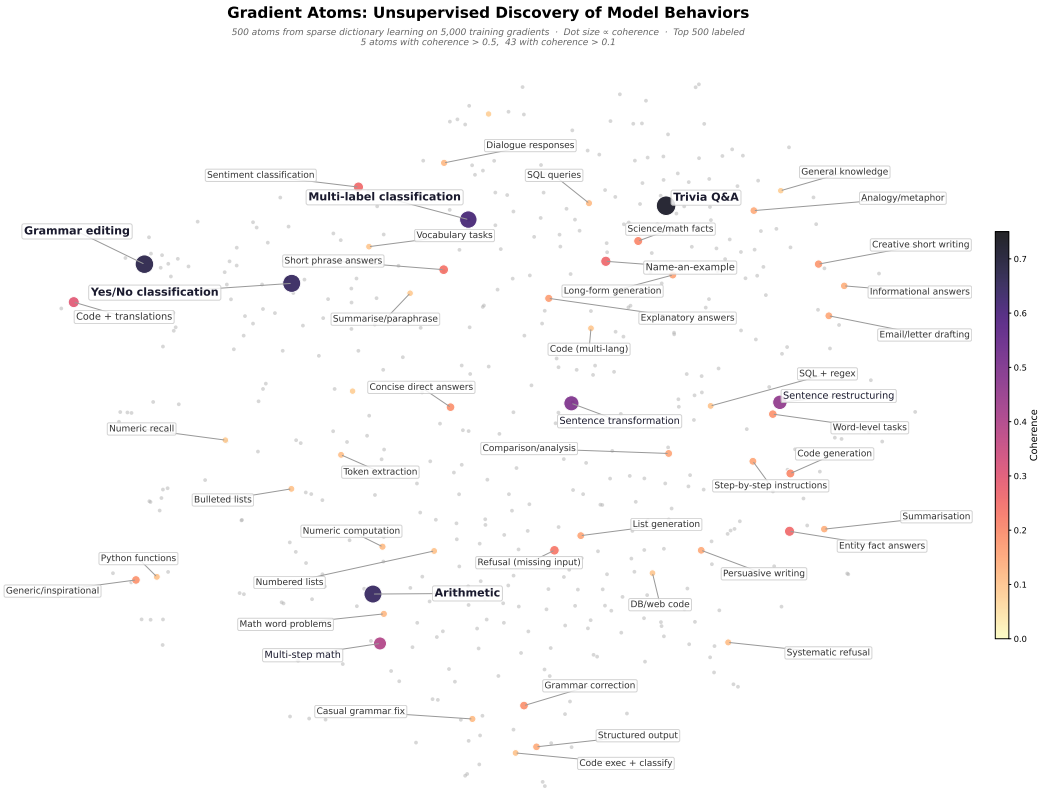

Figure 1: Gradient atoms discovered via sparse dictionary learning over 5,000 training-document gradients. Each point is one atom; high-coherence atoms correspond to tightly defined task types.

This is the idea behind **Gradient Atoms**. We extract per-document gradients, project them into a preconditioned eigenspace (via EKFAC), and apply sparse dictionary learning to decompose them into atoms. Each atom is a direction in weight space shared by a cluster of functionally similar documents. The method is:

- **Unsupervised**: no behavioral labels, measurement functions, or contrastive pairs are needed.
- **Efficient**: one decomposition surfaces candidate behaviors simultaneously, with no per-query scoring stage.
- **Actionable**: each atom can be unprojected into a full-parameter steering vector.

We validate this in two stages. First, we show that sparse dictionary learning on EKFAC-preconditioned gradients discovers 500 atoms, of which the highest-coherence atoms correspond to interpretable task types (Table 4). Second, we show that these atoms—discovered without any behavioral labels—function as effective **steering vectors**: applying them as weight perturbations produces large, controllable shifts in five tested behaviors (Figure 2).

Our contributions are:

1. We identify a limitation of per-document TDA and propose an alternative: decomposing the gradient space into shared update directions.
2. We introduce Gradient Atoms, an unsupervised method that discovers candidate model behaviors from training gradients alone, without supervision or per-query scoring.
3. We demonstrate that discovered atoms function as effective steering vectors, producing large controllable shifts in model behavior without any behavioral labels.

2 Related Work

Several concurrent works share our core insight that gradient similarity reflects functional similarity among training examples. GradientSpace (Sridharan et al., 2025) clusters LoRA gradients to identify "latent skills" but uses them to train specialised expert routers, not for interpretability or steering. Mode-Conditioning (Wu et al., 2025) confirms that gradient clustering reliably recovers functional groupings (98.7% F1), but applies this to test-time compute allocation. ELROND (Skierś et al., 2026) is philosophically closest: they decompose per-sample gradients into steerable directions via PCA and sparse autoencoders, but in diffusion models rather than LLMs.

The gap we fill is the combination of: (1) operating on training gradients rather than inference activations, (2) being fully unsupervised with no per-query scoring, and (3) the discovered atoms directly functioning as steering vectors for LLMs. Standard TDA methods (Koh & Liang, 2017; Grosse et al., 2023; Bae et al., 2024) are all supervised, requiring $O(Q \times N)$ query–document comparisons. In the activation-space literature, SAEs (Nanda et al., 2023) and gradient-informed variants (Olmo et al., 2025; Shu et al., 2025) decompose activations at inference time; atoms decompose what the model *learned* during training, across all layers simultaneously. Extended discussion is in Section A.

3 Method

Consider a language model with parameters θ fine-tuned on a dataset of N input–output pairs $\{x_1, \ldots, x_N\}$. Each document x_i induces a gradient $g_i = \nabla_\theta \mathcal{L}_{\text{CE}}(\theta; x_i) \in \mathbb{R}^d$, where d is the number of trainable parameters. The gradient g_i is the direction the model's weights would move to improve on document i. Gradient Atoms decomposes these per-document gradients into sparse components; the highest-quality components isolate interpretable learned behaviors. The pipeline has five steps.

3.1 Per-Document Gradient Extraction

For each training document x_i, we compute the gradient of cross-entropy loss with respect to all trainable parameters, producing a gradient matrix $G \in \mathbb{R}^{N \times d}$. Documents that require similar computations tend to produce similar gradient vectors.

3.2 EKFAC Projection and Preconditioning

The raw gradient space is anisotropic—some directions have high curvature (small weight changes cause large loss changes) and others have low curvature. Without correction, any decomposition is dominated by high-curvature directions, drowning out semantic structure. We use the EKFAC eigendecomposition (Grosse et al., 2023; Bae et al., 2024) of the approximate Fisher information matrix to correct for this. For each module m with eigenvectors Q_m and eigenvalues λ_m, we project each gradient into the top-k eigenvectors and precondition:

$$\tilde{g}_i^{(m)} = Q_m^{(k)\top} g_i^{(m)}, \qquad \hat{g}_i^{(m)} = \frac{\tilde{g}_i^{(m)}}{\sqrt{\lambda_m^{(k)} + \epsilon}} \tag{1}$$

After concatenating across all M modules, the projected gradient is $\hat{g}_i \in \mathbb{R}^{k_{\text{total}}}$ where $k_{\text{total}} = k \times M$. This makes the space approximately isotropic: a unit step in any direction corresponds to a roughly equal change in loss, encouraging atoms to capture functionally distinct directions rather than curvature artifacts.

3.3 Sparse Dictionary Learning

We normalize each projected gradient to unit norm (so atoms reflect direction, not magnitude) and apply sparse dictionary learning to decompose:

$$\hat{g}_i \approx \sum_{j=1}^{K} \alpha_{ij}\, d_j \tag{2}$$

where $D = [d_1, \ldots, d_K] \in \mathbb{R}^{K \times k_{\text{total}}}$ are the **atoms** and α_{ij} are sparse coefficients—most are zero, so each document is explained by a few atoms. The sparsity penalty encourages each atom to capture a single pattern rather than blending multiple unrelated behaviors.

3.4 Coherence Scoring

For each atom j, we identify its activating documents—those with non-zero coefficient α_{ij}—and compute a coherence score over the top-n activating documents $\mathcal{S}_j$:

$$\text{coherence}(j) = \frac{1}{|\mathcal{S}_j|(|\mathcal{S}_j| - 1)} \sum_{a \neq b \in \mathcal{S}_j} \cos(g_a, g_b) \tag{3}$$

where g_a, g_b are the *raw* (unprojected, full d-dimensional) gradients. High coherence suggests the atom has found a shared computational motif in the original weight space, rather than an artifact of the projection.

3.5 Unprojection to Steering Vectors

Any atom can be converted back to a full parameter-space vector by reversing the projection:

$$v_j = \text{unproject}(d_j) \in \mathbb{R}^d \tag{4}$$

This vector can be applied as a weight-space perturbation $\theta_{\text{new}} = \theta \pm \alpha \cdot v_j$, analogous to curvature-aware model editing (Ikram et al., 2026). The key difference is that v_j was discovered unsupervised from the training data, rather than derived from a hand-crafted measurement function or contrastive pair.

4 Experiments

4.1 Setup

Model. We use **Gemma-3 4B IT** fine-tuned via LoRA (rank 8) on the `q_proj` and `v_proj` matrices across all 34 layers, yielding 2.2M trainable parameters across 136 LoRA modules.

Dataset. The model is fine-tuned on 5,000 instruction–response pairs sampled from a general-purpose SFT mixture covering arithmetic, grammar correction, classification, code generation, QA, creative writing, and other tasks. EKFAC factors (eigendecomposition of the approximate Fisher) are computed on the full training set.

Gradient Atoms pipeline. We extract per-document gradients for all 5,000 training examples, project via EKFAC into 6,800 dimensions (50 eigencomponents $\times$ 136 modules, a $328\times$ reduction), and run MiniBatchDictionaryLearning (scikit-learn) with $K = 500$ atoms and sparsity penalty $\alpha = 0.1$. Coherence is computed over the top-20 activating documents per atom using the raw 2.2M-dimensional gradients.

4.2 Atom Discovery

From 500 atoms: 5 have coherence > 0.5, 43 have coherence > 0.1, and 457 have coherence < 0.1. Figure 1 shows the distribution. The top 5 atoms (coherence > 0.5) are: short factual QA (0.725), grammar editing (0.672), yes/no classification (0.647), simple arithmetic (0.643), and multi-category classification (0.614). The full top 50 are listed in Table 4.

Atoms capture task types, not topics. The decomposition clusters training data by *how* the model responds (arithmetic, classification, editing, code) rather than *what* it responds about (science, history, culture). This is consistent with Ruis et al. (2024), who find that procedural knowledge is a primary driver of what models extract from training data.

High coherence = stereotyped tasks. The top-5 atoms are all formulaic task types whose activating documents have highly similar gradients, suggesting similar computational pathways.

Multiple granularities. Grammar correction appears three times (ranks 2, 17, 36) and code generation five times (ranks 16, 37, 40, 45, 46), at decreasing coherence. The dictionary finds sub-clusters that may reflect different sentence complexity levels or programming language families.

Format atoms. Bulleted lists (#469) and numbered lists (#299) are separate atoms, suggesting the model uses distinct weight pathways for these formatting patterns.

Refusal is discoverable. Two atoms (#52, #161) capture the model's tendency to reply "Please provide the input" when task instructions lack content—a behavior learned from training data that appears to be separable from other behaviors in gradient space.

Effect of sparsity penalty. Table 1 shows the effect of α. At $\alpha = 0.01$, atoms are too dense (median $\sim$2500 docs each), blending unrelated patterns. At $\alpha = 0.1$, atoms are selective ($\sim$100 docs each). At $\alpha = 1.0$, the penalty overwhelms reconstruction and all coefficients are zero.

Table 1: Effect of sparsity penalty α on atom quality.

α	Docs per atom (median)	Atoms coh > 0.5	Atoms coh > 0.1
0.01	$\sim$2500	3	$\sim$20
0.1	$\sim$100	5	43
1.0	0	0	0

4.3 Behavioral Steering

A key test of whether gradient atoms capture genuine computational structure is whether they can **steer model behavior** when applied as weight-space perturbations. We select five atoms spanning a range of coherence scores and behavioral types, unproject each to a full LoRA parameter-space vector, and apply perturbations $\theta_{\text{new}} = \theta \pm \alpha \cdot v_j$ with $\alpha \in \{0.5, 1.0, 2.0, 5.0, 10.0\}$ in both directions. Since dictionary learning assigns atom signs arbitrarily, we test both. For each atom, we design 100 evaluation questions ($\sim$60 that naturally invite the target behavior, $\sim$40 neutral controls) and measure behavior with a regex detector:

- **Yes/No** (#415): first line starts with Yes/No/True/False
- **Code** (#64): response contains a fenced code block (```)
- **Refusal** (#161): matches clarification-seeking patterns
- **Bullets** (#469): ≥ 2 lines starting with $-$, $\star$, or $\bullet$
- **Numbered** (#299): ≥ 2 lines starting with `\d+[.)]`

Each steered adapter is served via vLLM alongside the clean baseline, with all 11 variants (5 alphas $\times$ 2 signs + baseline) loaded simultaneously as LoRA modules.

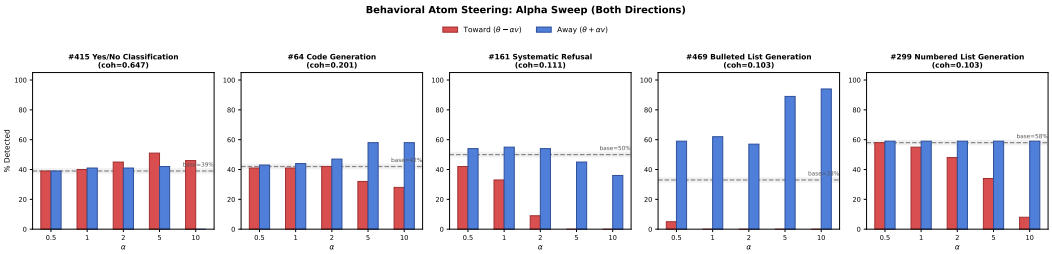

Figure 2: Behavioral steering via unsupervised gradient atoms. Red bars show the "toward" direction ($\theta - \alpha v$), blue bars show "away" ($\theta + \alpha v$), and the dashed line marks the clean baseline. Four of five atoms produce large, monotonic steering effects in at least one direction.

Results are shown in Figure 2 and Table 2. All five atoms steer behavior in at least one direction, and four produce large effects (>14pp):

Table 2: Behavioral atom steering results. For each atom, we report the baseline detection rate and the best increase/decrease achieved across all alpha values and both directions. Δ is the change in percentage points from baseline.

Atom	Behavior	Coh.	Base	Best ↑	Δ↑	Best ↓	Δ↓
#415	Yes/No Classification	0.647	39%	51%	+12pp	0%	−39pp
#64	Code Generation	0.201	42%	58%	+16pp	28%	−14pp
#161	Systematic Refusal	0.111	50%	55%	+5pp	0%	−50pp
#469	Bulleted Lists	0.103	33%	94%	+61pp	0%	−33pp
#299	Numbered Lists	0.103	58%	59%	+1pp	8%	−50pp

- **Bulleted lists (#469)** is the strongest: steering increases bullet usage from 33% to 94% (+61pp) and suppresses it to 0%. The effect is monotonic with alpha in both directions.
- **Refusal (#161)** is completely suppressible: from 50% baseline to 0% at $\alpha = 5$. The steered model responds "Okay." to underspecified prompts instead of asking for clarification. The reverse direction modestly increases refusal (+5pp) and makes the model more verbose, suggesting this atom captures a terse/verbose dimension.
- **Code generation (#64)** increases from 42% to 58% (+16pp) or decreases to 28% (−14pp). The sign is flipped relative to the Newton step convention, confirming that atom signs from dictionary learning are arbitrary.
- **Yes/No (#415)** shows moderate amplification (+12pp) but strong suppression (−39pp to 0%). At high alpha, the model loses coherence, establishing the upper bound of useful perturbation.
- **Numbered lists (#299)** is asymmetric: easily suppressed (58% → 8%) but not amplifiable (+1pp), likely due to a ceiling effect.

5 Discussion

Suppression appears easier than amplification. All five atoms suppress their target behavior to near zero, but only two (bullets, code) achieve substantial amplification. One interpretation is that suppressing a behavior requires disrupting a single computational pathway, while amplifying requires strengthening it against many competing alternatives.

Coherence does not necessarily predict steerability. Atom #469 (coherence 0.103) produces the largest steering effect (+61pp), while #415 (coherence 0.647) gives only +12pp. Coherence measures gradient alignment among activating documents, but steerability also depends on how much the model's default behavior already saturates the target pathway.

Limitations. Only 43 of 500 atoms have coherence > 0.1, the majority are noise or capture overly broad mixtures. The instruction-following training data means atoms recover task types rather than fine-grained semantic preferences; more naturalistic data might yield different atoms. The 6,800-dim EKFAC projection discards information, and 5,000 documents may not cover rare behaviors. Our regex-based evaluation measures surface formatting rather than deeper behavioral changes.

6 Conclusion

We presented Gradient Atoms, an unsupervised method for discovering what a fine-tuned model has learned by decomposing training gradients into sparse components. The method reframes training data attribution: instead of scoring individual documents against a known behavior, it discovers shared update directions that clusters of documents jointly induce. The highest-coherence atoms recover interpretable task-type behaviors without supervision, and function as effective steering vectors, connecting unsupervised behavior discovery with controllable model editing.

Future directions include composing multiple atoms for simultaneous multi-behavior steering, scaling the dictionary to 1,000+ atoms, cross-model comparison to identify shared vs. adapter-specific behaviors, and developing principled methods for alpha selection.

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A Extended Related Work

Training data attribution. Influence functions (Cook et al., 1982; Koh & Liang, 2017) estimate the effect of removing a training example on predictions. Grosse et al. (2023) scaled these to LLMs via EKFAC, and Bae et al. (2024) improved accuracy with approximate unrolled differentiation. All such methods are supervised—requiring a query behavior and $O(N)$ scoring per query—which Gradient Atoms avoids entirely.

Gradient-based clustering. GradientSpace (Sridharan et al., 2025) clusters LoRA gradients via online SVD to identify "latent skills," sharing our core insight that gradient similarity reflects functional similarity. However, they use clusters to train specialised LoRA experts with a router, not for interpretability or steering. Their SVD + k-means yields routing labels; our sparse dictionary learning produces individually steerable atoms. Mode-Conditioning (Wu et al., 2025) independently confirms that gradient clustering recovers functional groupings (98.7% F1), applying this to test-time compute allocation.

Gradient decomposition in diffusion models. ELROND (Skierś et al., 2026) decomposes per-sample gradients via PCA and sparse autoencoders into steerable directions for visual attribute control—the diffusion-model analogue of our approach. Key differences: they decompose gradients *within* a single prompt's realisations rather than across the full training set, and target visual attributes rather than LLM behaviors.

Activation-space interpretability. SAEs decompose single-layer activations into monosemantic features (Nanda et al., 2023); gradient-informed variants (g-SAEs (Olmo et al., 2025), GradSAE (Shu et al., 2025)) use gradients to improve feature selection. Both still decompose activations—we decompose the gradients themselves, across all layers simultaneously. Wang & Murfet (2026) frame the theoretical dual of interpretability (behavior $\rightarrow$ training causes); Gradient Atoms provides a practical mechanism that additionally discovers behaviors without a query.

Model editing and procedural knowledge. Prior steering methods require a known concept with measurement functions or contrastive pairs (Ikram et al., 2026); Gradient Atoms discovers steering directions unsupervised. Our finding that atoms capture task types rather than topics is consistent with Ruis et al. (2024), who show procedural knowledge drives what models extract from training data.

B Computational Details

Table 3: Computational cost of the Gradient Atoms pipeline.

Step	Resources	Time
Gradient extraction	$8\times$ A100 40GB	170s
EKFAC projection	CPU, 16GB RAM	$\sim$5 min
Dictionary learning ($\alpha = 0.1$)	CPU, 32GB RAM	$\sim$15 min
Coherence computation	CPU, 8GB RAM	$\sim$5 min
Total		**$\sim$25 min**

Model: Gemma-3 4B IT, LoRA rank 8 (`q_proj` + `v_proj`), 2.2M parameters, 136 modules across 34 layers. EKFAC factors computed separately on the full training set.

C Full Atom Table

Table 4: Top 50 gradient atoms ranked by coherence score. Each atom was characterised by manual inspection of its top-20 activating documents.

Rank	Atom	Coherence	Active Docs	Description
1	#348	0.725	139	Short factual Q&A—trivia with one-word/numeric answers
2	#328	0.672	110	Grammar and sentence editing
3	#415	0.647	156	Yes/No/True/False binary classification
4	#458	0.643	124	Simple arithmetic
5	#498	0.614	176	Multi-category classification and labeling
6	#358	0.499	88	Sentence transformation (voice, tense, translation)
7	#2	0.463	206	Sentence restructuring (questions, passive/active)
8	#451	0.395	182	Multi-step arithmetic and unit conversions
9	#484	0.298	49	Mixed technical (code + translations + set ops)
10	#319	0.262	180	“Name an example of X”—single-entity retrieval
11	#430	0.258	150	Sentiment and text classification
12	#425	0.257	215	Single-entity factual answers
13	#363	0.238	146	Short phrase answers to open questions
14	#52	0.230	57	“Please provide the input”—refusal on missing input
15	#364	0.205	158	Science and math fact answers
16	#64	0.201	25	Code generation (Python, JS, C++, HTML)
17	#303	0.189	49	Grammar correction on short sentences
18	#394	0.188	144	Concise direct answers (mixed tasks)
19	#488	0.187	168	Short inspirational/generic responses
20	#477	0.185	227	Word-level tasks (synonyms, antonyms, rhymes)
21	#376	0.176	97	Creative short-form writing
22	#136	0.165	161	Multi-sentence explanatory answers
23	#66	0.154	45	Long-form generation (essays, paragraphs)
24	#457	0.152	83	Comparison and analysis tasks
25	#256	0.152	50	Step-by-step instructions and how-to guides
26	#265	0.151	118	List generation (brainstorming, idea lists)
27	#224	0.149	31	Email and letter drafting
28	#446	0.146	86	Persuasive/argumentative writing
29	#294	0.142	50	Data extraction and structured output
30	#419	0.142	78	Analogy and metaphor reasoning
31	#359	0.137	211	Neutral informational answers
32	#306	0.136	118	Summarisation
33	#181	0.119	87	Dialogue and conversational responses
34	#72	0.118	37	Math word problems
35	#445	0.117	69	Numeric computation (GCF, LCM, time)
36	#465	0.116	70	Grammar correction on casual sentences
37	#231	0.115	21	SQL queries and structured code
38	#161	0.111	47	Systematic refusal on unclear input
39	#325	0.106	143	Single-word/token extraction from input
40	#61	0.105	21	Python utility function implementations
41	#469	0.103	143	Bulleted list generation
42	#299	0.103	46	Numbered list generation
43	#67	0.102	9	SQL + regex + technical expressions
44	#180	0.100	52	Mixed code execution and classification
45	#381	0.097	79	Code generation (broad, multi-language)
46	#428	0.096	81	Database/web code (SQL, HTML, CSS, APIs)
47	#48	0.095	56	Single-word vocabulary tasks (fill-in-blank, plurals)
48	#475	0.088	83	Summarisation and paraphrasing
49	#233	0.087	46	Numeric/factual recall with approximation
50	#172	0.084	129	General knowledge Q&A